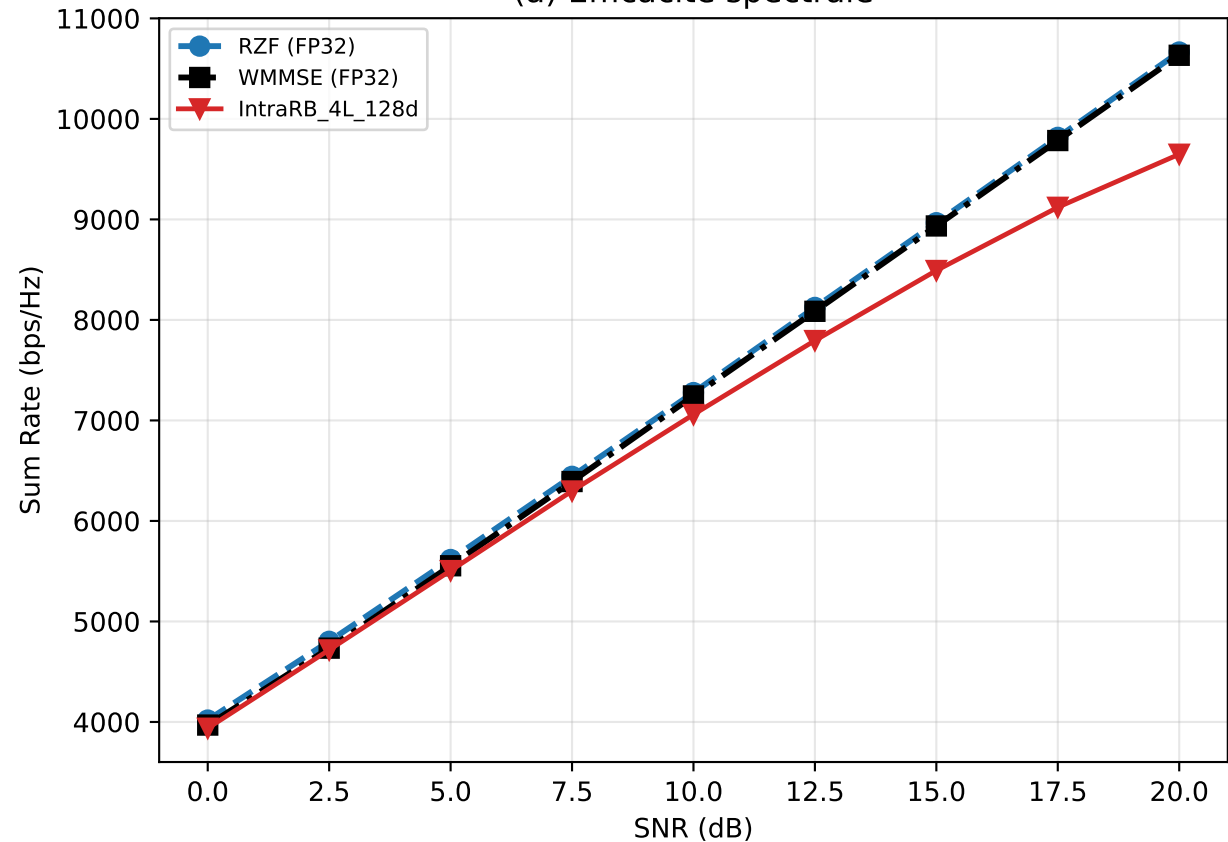
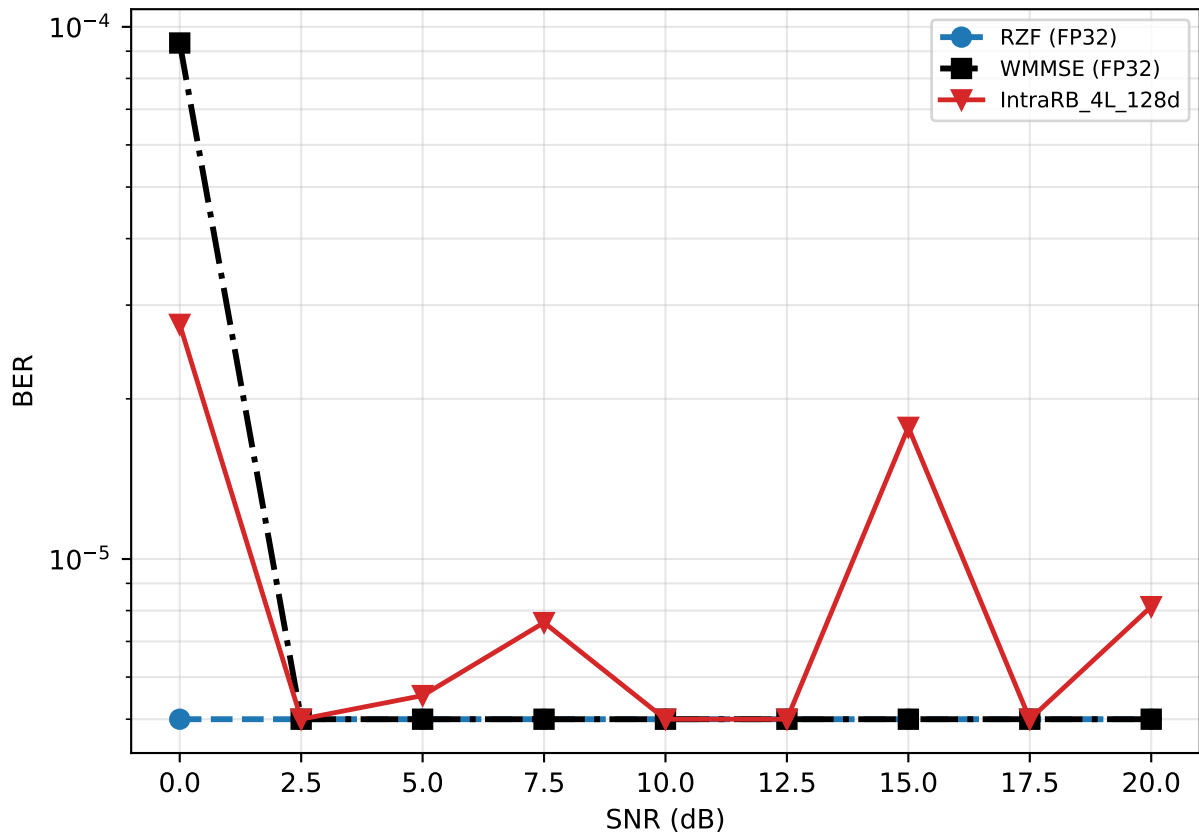


IntraRB Transformer Precoder — 8×4 MU-MIMO (8TX, 4UE, 72SC, SNR train=15dB)

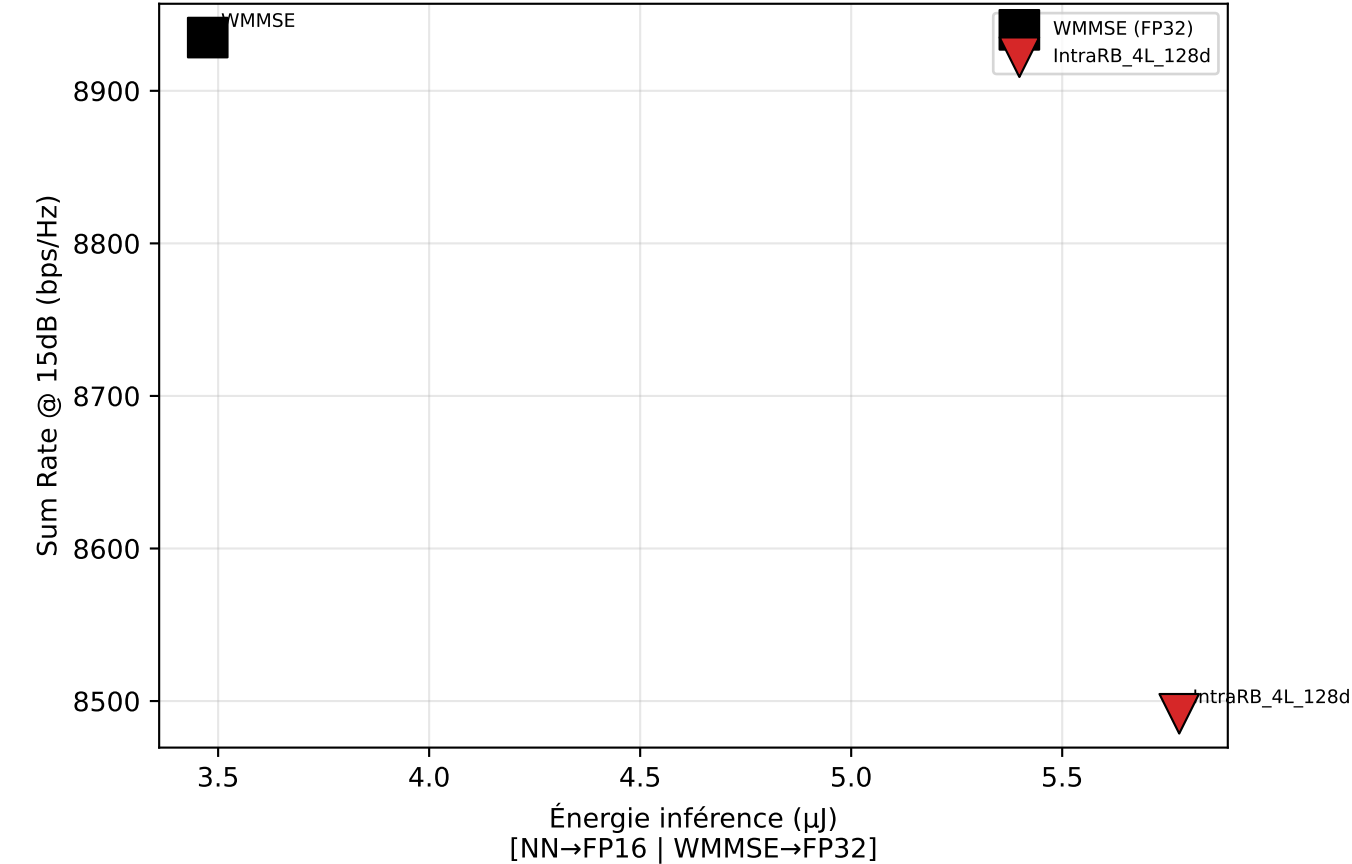
(a) Efficacité spectrale



(b) BER Monte Carlo



(c) Pareto Énergie-Performance



(d) Résumé @ 15dB

Méthode	Rate (bps/Hz)	Gap vs WMMSE	FLOPs (M)	E FP16 (μ J)	Gain FP16
RZF (FP32)	8967.23	+32.33	2	N/A	ref
WMMSE (FP32)	8934.90	+0.00	1842	N/A	ref
IntraRB_4L_128d	8491.66	-443.25	10695	5.7773	0.6×